Project Planning & Management

Project Proposal

Overview: This project focuses on the design and implementation of a Mini Security Information and Event Management (SIEM) system that leverages Artificial Intelligence (AI) to autonomously detect suspicious and anomalous activities within network traffic and system logs.

Objectives:
Develop an anomaly-based detection system capable of identifying unknown or zero-day attacks without prior knowledge of attack patterns.

Utilize unsupervised machine learning models to learn "normal" system behavior and flag deviations as potential threats.
Reduce false positive rates to mitigate alert fatigue for security analysts.

Scope: The system ingests real-time network traffic logs, applies multiple detection models (Isolation Forest, OC-SVM, and Autoencoders), and visualizes alerts via an interactive dashboard.

Project Plan (Milestones & Deliverables)

Milestone 1: Data Collection & Preprocessing: Focuses on gathering network traffic data and engineering features relevant to security analysis. Deliverable: Cleaned and processed log dataset.

Milestone 2: Model Development: Building and evaluating unsupervised learning models like Isolation Forest and Autoencoders. Deliverable: Trained models and comparative performance report.
Milestone 3: Dashboard & Alert System: Developing a real-time Streamlit dashboard and an AlertManager to categorize threats. Deliverable: Functional Streamlit Command Center.

Milestone 4: Advanced Features: Implementing ensemble learning, concept drift detection, and Explainable AI (XAI). Deliverable: Security Copilot module for alert explanation.
Milestone 5: Final Documentation: Creating automated incident reports and the final technical report. Deliverable: Final Technical Report and System Demo.

Task Assignment & Roles

Nessma Mostafa AbdElrahiem:Team Leader & ML Lead; responsible for overall system architecture, model selection (Ensemble), and code integration.
Ahmed Sabry Mohamed: Data Engineer; responsible for cleaning datasets, feature engineering, and developing the attack simulation scripts.
Mahmoud Ahmed Salama: Documentation Lead; responsible for generating automated PDF reports and maintaining the technical documentation.
Fady Melad Zarif: Frontend Developer; responsible for creating the Streamlit dashboard and real-time visualization components.
Maryam Helal Mahmoud: Logic Developer; responsible for the AlertManager module and defining the thresholds for High/Medium/Low alerts.
Rawan Mohamed Mostafa: Data Analyst; responsible for Exploratory Data Analysis (EDA) and implementing the statistical drift detection module.

Risk Assessment & Mitigation Plan

Risk: High false positive rates in unsupervised models.
Mitigation: Implementing a hybrid classification strategy and conservative thresholding (e.g., 95th percentile) to prioritize precision.

Risk: Missing stealthy "low and slow" attacks.
Mitigation: Using ensemble learning to combine votes from multiple models for higher detection sensitivity.

Key Performance Indicators (KPIs)

Detection Accuracy: Precision and Recall metrics on synthetic attack datasets.

Response Time: Latency between log ingestion and dashboard alert notification.

Alert Efficiency: Reduction in redundant alerts through the intelligent deduplication mechanism.